Example Project 1: Are Moscow's Januaries Wetter Than They Used to Be?

EXAMPLE STUDENT REPORT (CATEGORICAL VARIABLE), STAT 251 SECTION 01

> **INSTRUCTOR NOTE**
>
> This is a completed example of the course project. It shows what a strong report looks like when the variable being analyzed is **categorical**. It follows the six steps on the Project Description (project.html) page in order, so you can compare it against the rubric section by section.
>
> The two example projects deliberately come from very different fields: this one uses weather records, and Example Project 2 (example_project_continuous.html) uses consumer lending. Any topic works as long as the data can be cited and your question fits a method from class. Use these examples to see the level of detail and explanation expected. **Do not copy their datasets, questions, or wording.**
>
> The example was prepared with the help of Claude (Anthropic), and every calculation in it was done with the course's stat-tools (https://stat-tools.jmks-homelab.dev/) site. The screenshots come from those tools.

1. Dataset Selection

(A) Citation.

> Lawrimore, J. H., Ray, R., Applequist, S., Korzeniewski, B., & Menne, M. J. (2016). *Global Summary of the Month (GSOM), Version 1* [Data set]. Station USC00106152, MOSCOW U OF I, ID US, monthly total precipitation (PRCP), January 1895 – January 2026. NOAA National Centers for Environmental Information. https://doi.org/10.7289/V5QV3JJ5. Accessed September 24, 2026.

Working link to the exact data I downloaded (one row per month, standard units): NCEI Access Data Service query for station USC00106152 (https://www.ncei.noaa.gov/access/services/data/v1?dataset=global-summary-of-the-month&stations=USC00106152&startDate=1890-01-01&endDate=2026-02-28&dataTypes=PRCP&units=standard&format=csv&includeAttributes=true&includeStationName=true). The station's record page is GHCND:USC00106152 (https://www.ncei.noaa.gov/cdo-web/datasets/GHCND/stations/GHCND:USC00106152/detail).

(B) Background.

Topic and station. The data describe precipitation (rain plus melted snow) at the weather station of the **University of Idaho experiment station**, about 2 miles east of Moscow. NOAA's station records give its location as 46.728° N, 116.956° W, at an elevation of 2,634 ft (802.8 m). The station's daily record in NOAA's archive begins on November

7, 1893 (station record (https://www.ncei.noaa.gov/cdo-web/datasets/GHCND/stations/GHCND:USC00106152/detail); station history (https://www.ncei.noaa.gov/access/homr/services/station/search?qid=COOP:106152&headersOnly=false)).

Who collects the data, and how.

- **The program.** This is a National Weather Service **Cooperative Observer Program (COOP)** station. The COOP program was formally created in 1890, and today more than 8,700 volunteers record the daily temperature and 24-hour precipitation (NWS COOP overview (https://www.weather.gov/coop/Overview)).
- **The gauge.** Precipitation is measured in the **Standard 8-inch Rain Gauge**. Its funnel sends the water into a narrow measuring tube whose cross-section is one-tenth the size of the funnel, so the water stands 10 times deeper than the rain that fell. That makes it possible to read the amount to the nearest **0.01 inch** (NWS: the 8-inch gauge (https://www.weather.gov/iwx/coop_8inch)).
- **Snow.** In winter, the funnel and tube are removed and snow collects in the outer can. It is then melted and measured as water, so a January total counts snow by its **water content**, not its depth.
- **Daily recording.** The observer reads the gauge once a day at a fixed time (currently 5 PM at this station) and records the 24-hour total:
  - 0.00 if nothing fell
  - "T" (trace) if there was too little to measure
  - "M" for a missed day, with the accumulated amount recorded when the gauge is next read

  (NWS Co-op Quick Reference Guide (https://www.weather.gov/media/mrx/coop_pdf/CooperativeWeatherObservationQuickReferenceGuide.pdf)).

How the monthly file is produced.

- **Archiving and monthly totals.** The daily reports are quality-checked and archived by NOAA's National Centers for Environmental Information (NCEI) in the Global Historical Climatology Network–Daily (GHCN-Daily). The **Global Summary of the Month (GSOM)** adds up each month's daily values to give the monthly total, `PRCP` (GSOM documentation (https://www.ncei.noaa.gov/pub/data/cdo/documentation/gsom_doc.pdf)).
- **The attribute field.** Each total comes with a `PRCP_ATTRIBUTES` field. Its second entry is a measurement flag:
  - `a` means the month's total includes missing days, so it is probably too low.
  - `T` means only a trace fell.

  Its last entry says which archive the value came from:
  - `0` means NOAA's Cooperative Summary of the Day dataset (DSI-3200), used for every month through 2010.
  - `7` means WxCoder, the online system that COOP observers use to submit their reports, used from 2011 on.

What the file contains. The file I downloaded has **1,574 rows**, one per month from December 1893 to February 2026, and five columns: `STATION`, `NAME`, `DATE`, `PRCP` (inches) and `PRCP_ATTRIBUTES`. In the whole file:

- 3 months carry the `a` flag: December 1893, February 1894 and January 1963.
- 18 carry `T`.

- 1 has no value at all.

How the original collectors sampled. They did not sample: the station records **every** day. So the file is a complete record of one location, not a random sample of places or days. I kept only the January rows, which gives **130 usable Januaries**:

- 94 “historical” Januaries from 1895 to 1990.
- 36 “recent” Januaries from 1991 to 2026.
- I dropped:
  - **1894**: no value.
  - **1927**: no record in the file.
  - **1963**: it carries the `a` flag, so its total is missing some days and would undercount that month.

2. Question and Hypothesis Formulation

(A) Question. Moscow gets much of its yearly precipitation in winter, and January is one of its wettest months. I want to know whether recent Januaries (1991–2026) are distributed across the dry-to-wet range the same way the historical Januaries (1895–1990) were. For example, is a “very wet” January more common now than it used to be?

(B) Variables.

- **January precipitation class** (**categorical**, ordinal). This is the variable I test. Each January is put in one of four classes using the *historical* quartiles of January total precipitation:
  - **Very dry:** below Q1.
  - **Dry:** between Q1 and the median.
  - **Wet:** between the median and Q3.
  - **Very wet:** above Q3.

  By construction, about one quarter of historical Januaries fall in each class. So if recent Januaries behave like historical ones, about one quarter of them should land in each class too.
- **January total precipitation** in inches (quantitative). This is the measurement the classes are built from. I describe it in Part 4, but I do not test it directly.

(C) Hypothesis in words. Recent Januaries fall into the four historical precipitation classes in equal proportions (25% each), the same as the historical record. The alternative is that at least one class is more or less common now than it used to be.

From a continuous measurement to ordered categories

A January’s total precipitation is a **quantitative, continuous** measurement. My question, though, is about *how often* Januaries are dry or wet compared with the past, not about the average amount. To answer it, I turned each total into an **ordered category**.

Step 1: choose the reference period. I set the cutoffs using only the **historical** Januaries (1895–1990) and never my sample. If I set the cutoffs from the same Januaries I am testing, they would split my sample evenly by construction, and the test could never find a difference.

Step 2: choose how many classes, and where to cut. I used the historical **quartiles**, which gives four classes that each contain about a quarter of historical Januaries.

- **Why equal-probability classes.** They make the null hypothesis simple ("25% in each class") and give every class the same expected count (Part 5).
- **Why four classes.** Four classes separate "very wet" from merely "wet", which is the difference my question is about.
- **The alternative.** NOAA's Climate Prediction Center uses **terciles** instead. It splits its 30-year reference period into the driest third, the middle third and the wettest third, called "below", "near" and "above" normal, each with a 1/3 chance (CPC outlook explanation (https://www.cpc.ncep.noaa.gov/products/predictions/erf_info.php?outlook=814&var=p)). I chose quartiles over terciles because they give one more level of detail at the extremes. With my sample size, the expected counts would still be large enough (24 × ¼ = 6 per class).

I entered the 94 historical January totals into the stat-tools **Statistic Calculator**, which computes quartiles with R's default (type 7) definition:

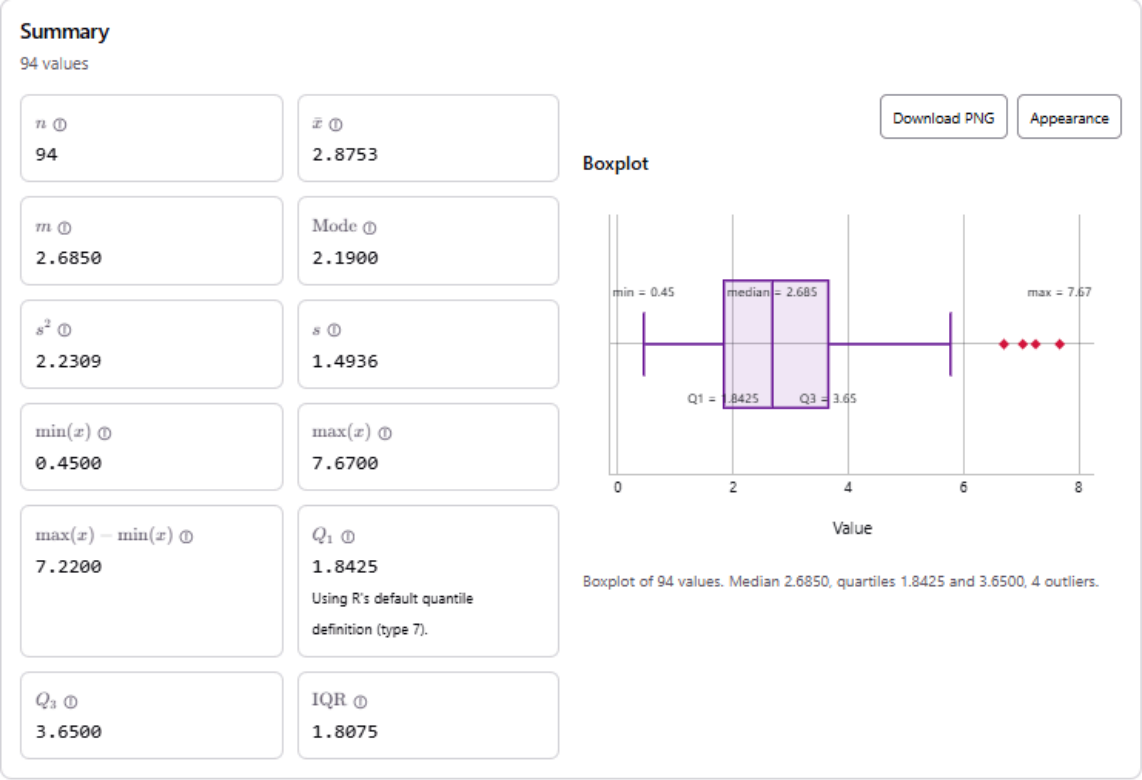


Figure 1. Statistic Calculator summary and boxplot of the 94 historical January totals (inches), 1895–1990. The quartiles set the class cutoffs.

This gives the cutoffs **Q1 = 1.8425 in**, **median = 2.685 in**, and **Q3 = 3.65 in**.

Step 3: fix the boundary rule. Every total must land in exactly one class, so I made each class include its lower cutoff and exclude its upper cutoff. For example, a January with exactly 3.65 in counts as "Very wet". Two historical Januaries fall exactly on Q3, but none of my sampled Januaries sit on any cutoff.

Table 1. Precipitation classes, defined from the 1895–1990 quartiles, and how many of the 94 historical Januaries fall in each.

Class	January total (inches)	Historical Januaries
Very dry	less than 1.8425	24 (25.5%)
Dry	1.8425 up to (not including) 2.685	23 (24.5%)
Wet	2.685 up to (not including) 3.65	22 (23.4%)
Very wet	3.65 or more	25 (26.6%)

The historical classes are close to, but not exactly, 25% each. There are two reasons: 94 cannot be split into four equal groups, and two Januaries sit exactly on the Q3 cutoff. I still use 25% as the benchmark, because that is what "the historical quartiles" means. The small unevenness is one of the limitations in Part 6.

Step 4: classify each sampled January. Here are three of my sampled years, classified step by step:

Year	January total	Comparison with the cutoffs	Class
2023	1.45 in	1.45 < 1.8425 (below Q1)	Very dry
2017	2.55 in	1.8425 ≤ 2.55 < 2.685 (between Q1 and the median)	Dry
2003	6.47 in	6.47 ≥ 3.65 (at or above Q3)	Very wet

What kind of variable this creates, and what it costs.

- **An ordinal variable.** The result is **categorical and ordinal**: Very dry < Dry < Wet < Very wet.
- **Information is lost.** A 3.66-inch January and a 6.87-inch January are both simply "Very wet". The categories keep *where* each January falls relative to history, but not *how far* inside its class it is.
- **Why that loss is acceptable here.** My question is about how often each kind of January happens, and I still report the actual totals in Part 4.

3. Sampling

(A) and (B) Method. My population is the **36 Januaries from 1991 to 2026**. I drew a **simple random sample of n = 24 Januaries** without replacement, using the stat-tools **Simple Random Sample** tool. To repeat it:

1. Save the 36 recent Januaries as a CSV with the columns `year_id` (1–36), `year` and `jan_prcp_in`.
2. Open the **Simple Random Sample** tool and upload the CSV.

3. Keep all variables, set **Sample size = 24** and **Seed = 251**, and press *Draw sample*. The same seed always draws the same 24 years.

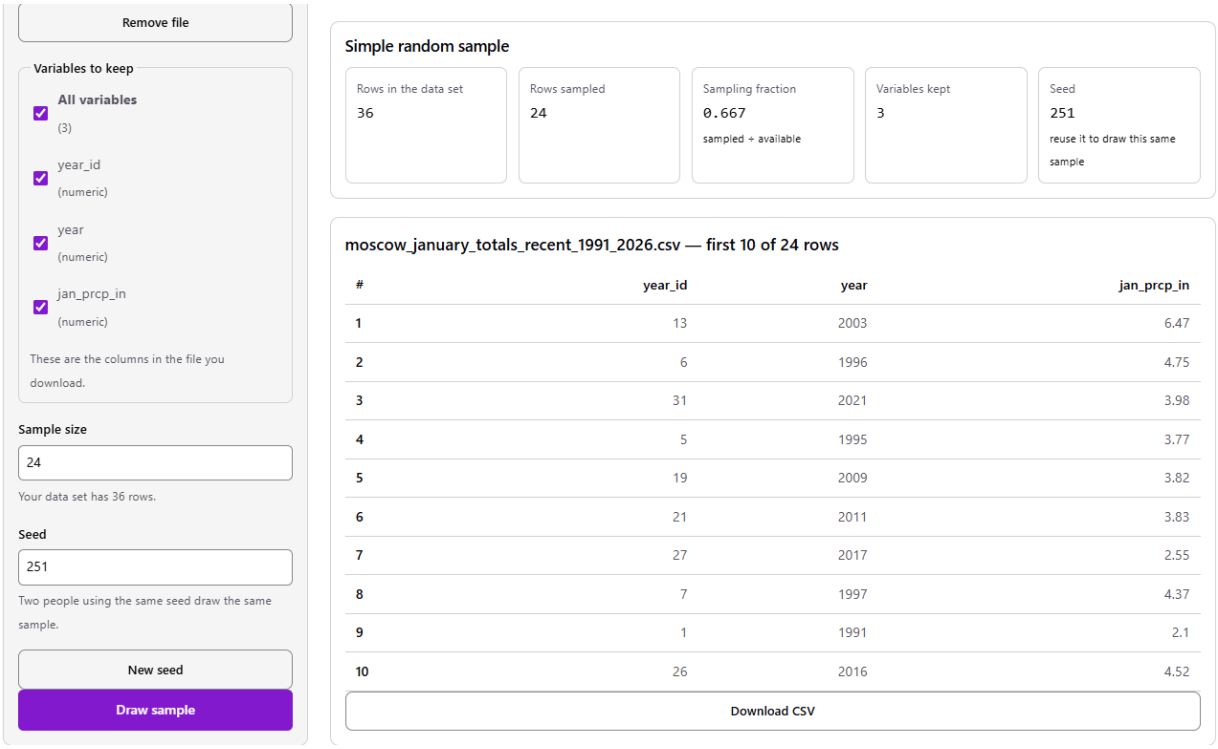


Figure 2. The Simple Random Sample tool after the draw: 24 of 36 rows sampled with seed 251.

(C) Justification.

- **Why SRS.** An SRS gives every recent January the same chance of being chosen. That means no year was picked because I remembered it as a big snow year.
- **Why years rather than days.** I sample whole years instead of individual days, and that is deliberate. Rainy days come in storm systems, so neighbouring days are not independent. Two different Januaries are much closer to independent.
- **Why n = 24.** 24 is within the required 20–30 and divides evenly into four classes. Under H_0 each class then expects $24 \times 0.25 = 6$ Januaries, which satisfies the rule of thumb that every expected count be at least 5 (Part 5).
- **Why the historical Januaries are not sampled.** They are not part of the sample. They play the role of the *benchmark* that defines the classes, the way a published value would in a one-sample test.

Possible bias. The SRS itself is unbiased. However, my population is small, so 24 of 36 is a large fraction (67%), and my conclusions only describe 1991–2026 at this one station.

A bigger worry is the station record itself, and no sampling method can fix it. NOAA's station history shows the station **moved 3.2 miles east-southeast in December 1960**. That falls inside my historical reference period, so Januaries before and after the move were measured at slightly different spots. If the two sites catch different amounts of precipitation, old and new totals are not perfectly comparable.

(D) The sample is listed in full in Appendix A.

4. Exploratory Statistical Description

(A) Frequency table. I classified each sampled January using Table 1 and counted the classes with the stat-tools **Frequency Tables** tool:

Table 2. Frequency table of the precipitation class for the 24 sampled Januaries.

Class	Frequency	Relative frequency
Very dry	2	0.083
Dry	8	0.333
Wet	3	0.125
Very wet	11	0.458
Total	24	1.000

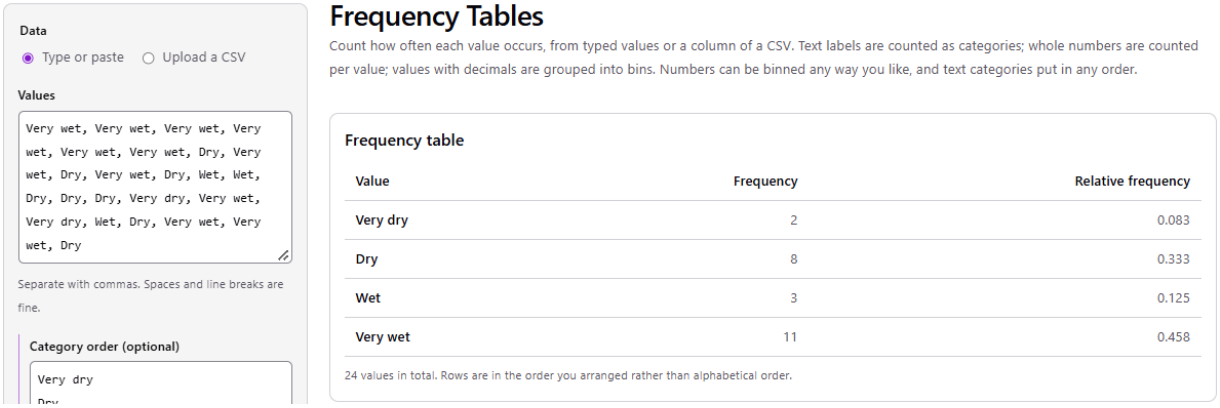


Figure 3. The Frequency Tables tool's output, which matches Table 2.

The **modal category is "Very wet"**: 11 of the 24 Januaries (45.8%) fall in it. Under the historical pattern only 25% of Januaries would.

(B) Plots.

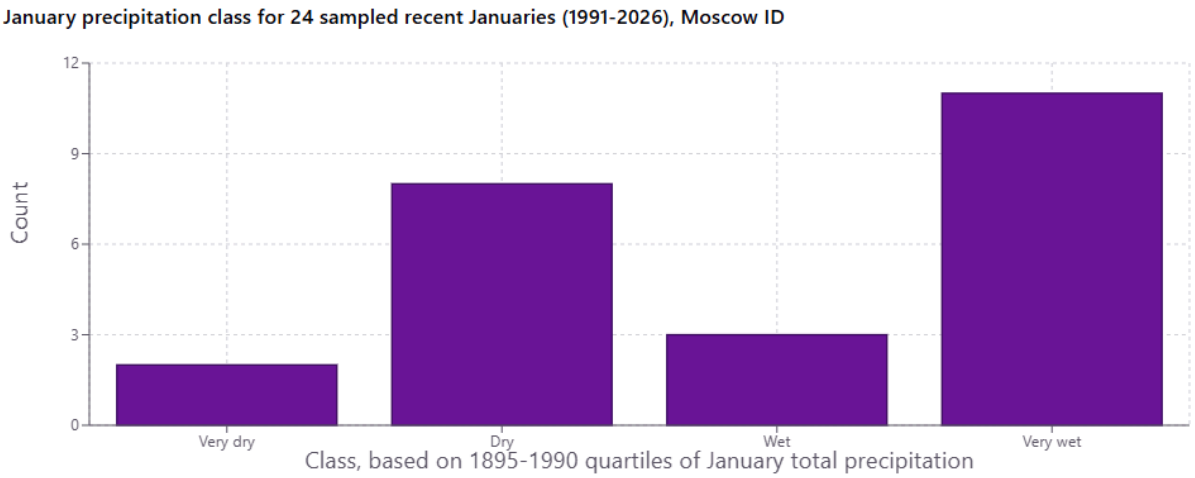


Figure 4. Bar chart (Make Graphs tool) of the precipitation class for the 24 sampled recent Januaries. If recent Januaries followed the historical pattern, each bar would be close to 6.

The class comes from a quantitative measurement, so I also summarized the sampled January totals themselves with the **Statistic Calculator**:

Table 3. Descriptive statistics of January total precipitation (inches) for my sample, alongside the historical reference values from Figure 1.

Statistic	Sample (1991–2026), n = 24	Historical (1895–1990), N = 94
minimum	1.4500	0.4500
Q1	2.3300	1.8425
median	3.2850	2.6850
mean	3.4229	2.8753
Q3	4.0775	3.6500
maximum	6.8700	7.6700
interquartile range	1.7475	1.8075
standard deviation	1.3952	1.4936

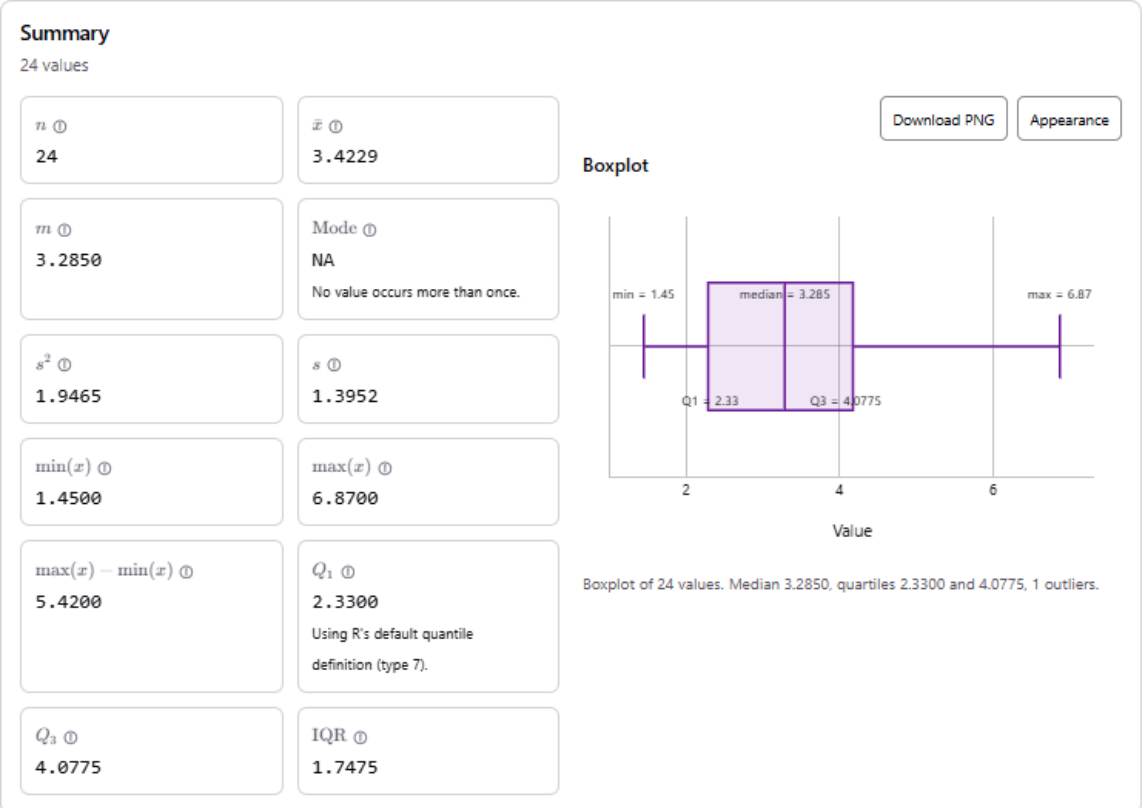


Figure 5. Statistic Calculator output and boxplot for the 24 sampled January totals (inches).

(C) Shape, center and variability.

- **Shape.** The class distribution is **not flat**. It piles up at the "Very wet" end (11) and the "Dry" class (8), while "Very dry" (2) and "Wet" (3) are rare.
- **Center and spread.** The underlying totals tell the same story. The sample's median January had 3.285 in, compared with 2.685 in historically, and its mean was 3.42 in versus 2.88 in. The spread is similar: the sample SD is 1.40 in and the historical SD is 1.49 in, and the IQRs are 1.75 in and 1.81 in.
- **Outliers.** January 2006 (6.87 in) lies above the upper fence $Q_3 + 1.5 \cdot IQR = 4.0775 + 1.5(1.7475) = 6.70$ in. The tool's caption also counts it as 1 outlier, although its boxplot draws the whisker all the way to 6.87. An outlier cannot distort a class count the way it can distort a mean: 2006 simply counts as one "Very wet" January. So I kept it.

5. Hypothesis Testing

(A) Hypotheses. Let p_{VD}, p_D, p_W, p_{VW} be the proportions of recent (1991–2026) Januaries in the Very dry, Dry, Wet and Very wet classes.

$$H_0 : p_{VD} = p_D = p_W = p_{VW} = 0.25$$

$$H_a : \text{at least one } p_i \neq 0.25$$

In words, H_0 says recent Januaries are spread across the historical classes exactly as historical Januaries were. H_a says at least one class is now more or less common. I use a significance level of $\alpha = 0.05$.

(B) Choice of test and assumptions. I use a χ^2 **goodness-of-fit test**. It compares the observed counts of a single categorical variable with the counts a stated distribution predicts, which is exactly my question. A one-proportion z test could only look at one class at a time. The assumptions are:

1. **Random sample.** Satisfied: the Januaries were chosen by SRS (Part 3).
2. **Independent observations.** Reasonable: different Januaries are a year apart, so one January's total tells you little about the next one's.
3. **Large enough expected counts.** Every expected count is $np_i = 24(0.25) = 6 \geq 5$. Satisfied.

(C) Computations.

Step 1: observed and expected counts. The expected count for each class is $E_i = np_i = 24 \times 0.25 = 6$.

Table 4. Worked calculation of the χ^2 statistic.

Class	Observed (O)	Expected (E = 24 × 0.25)	O − E	$(O - E)^2 / E$
Very dry	2	6	-4	2.667
Dry	8	6	2	0.667
Wet	3	6	-3	1.500
Very wet	11	6	5	4.167
Total	24	24	0	**9.000**

Step 2: test statistic.

$$X^2 = \sum_{i=1}^{4} \frac{(O_i - E_i)^2}{E_i} = \frac{(2-6)^2}{6} + \frac{(8-6)^2}{6} + \frac{(3-6)^2}{6} + \frac{(11-6)^2}{6} = \frac{16+4+9+25}{6} = \frac{54}{6} = 9.000$$

Step 3: degrees of freedom and critical value. There are $k = 4$ classes, so $df = k - 1 = 3$. The critical value is $\chi^2_{3,\,0.95} = 7.815$, and we reject H_0 if $X^2 \geq 7.815$.

Step 4: p-value. $p = P(\chi^2_3 \geq 9.000) = 0.0293$.

I ran the same test in the stat-tools **Categorical Tests** tool. I uploaded my sample CSV, chose the `jan_class` column, and set the expected proportions to 0.25, 0.25, 0.25, 0.25. The tool lists the classes alphabetically, and it reproduces every number above:

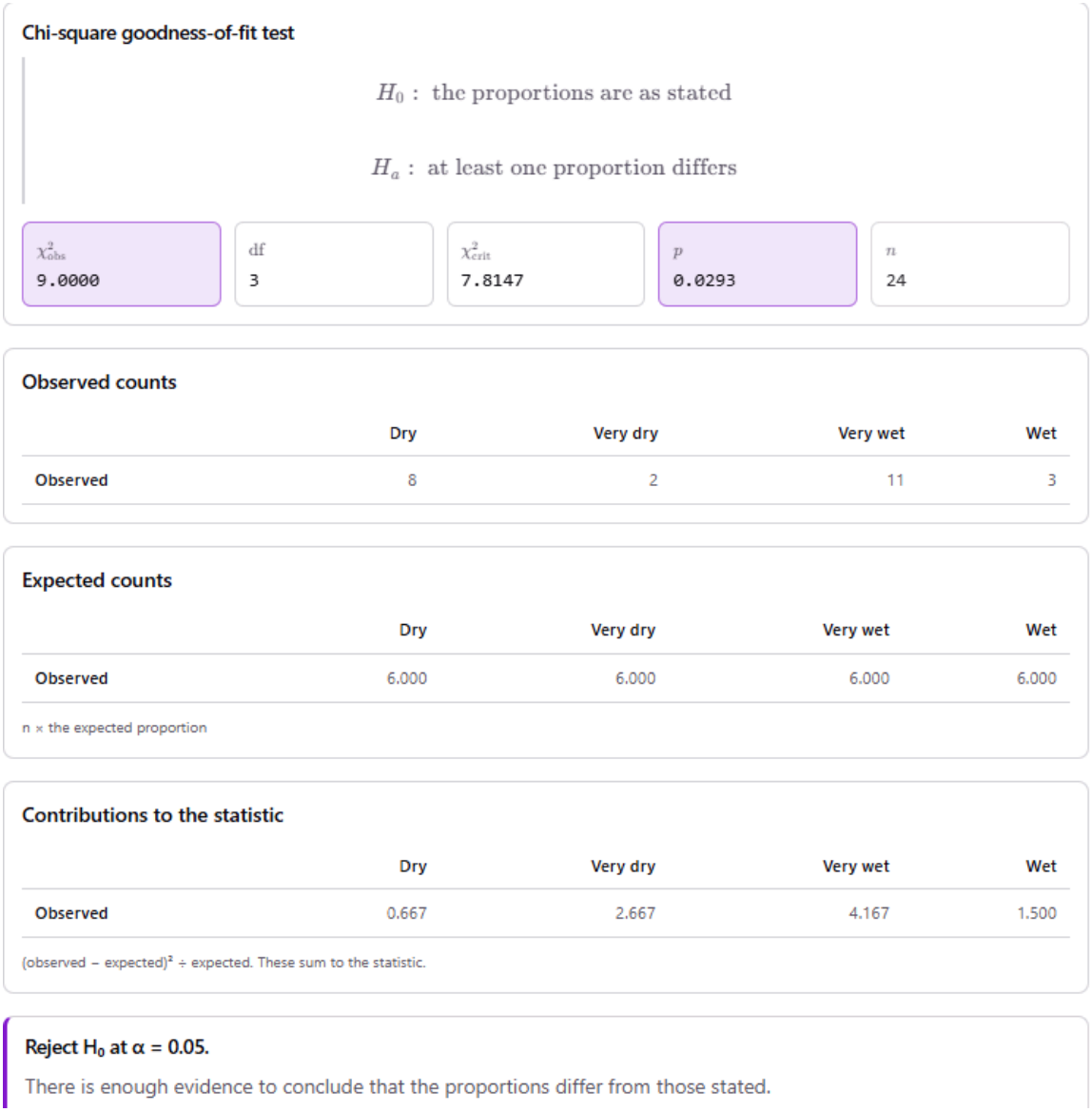


Figure 6. Categorical Tests tool output for the goodness-of-fit test: X^2 = 9.0000, df = 3, critical value 7.8147, p = 0.0293.

Quantity	Value
Test statistic X^2	9.000
Degrees of freedom	3
Critical value $\chi^2_{3,\,0.95}$	7.815
p-value	0.0293

6. Conclusion

(A) Decision. The test statistic $X^2 = 9.00$ is larger than the critical value 7.815. Equivalently, $p = 0.029 < \alpha = 0.05$. So I **reject** H_0.

(B) Interpretation. In plain language, recent Moscow Januaries do not fall into the dry-to-wet classes the way Januaries did from 1895 to 1990. The contributions in Table 4 show *where* the difference comes from:

- **"Very wet" contributes the most (4.17 of the 9.00).** Almost half of the sampled recent Januaries (11 of 24) were wetter than 75% of historical Januaries, when only about 6 would be expected.
- **"Very dry" contributes the next most (2.67).** Only 2 of the 24 were drier than 75% of historical Januaries, instead of about 6.

In other words, the data suggest that very wet Januaries have become more common in Moscow and very dry ones less common. The quantitative summaries agree: the sample's median January total is about 0.6 inches higher than the historical median.

One caution: the χ^2 test by itself only says the distribution *changed*. The "wetter" direction comes from looking at the table. The test does not tell us *why* the pattern changed.

(C) Limitations.

- **Small sample and a fixed benchmark.** With only 24 Januaries, the result rests on a handful of years. Also, the cutoffs are quartiles *estimated* from 94 historical Januaries, but the test treats them as known exactly.
- **Classes are only approximately 25%.** Historically the four classes held 24, 23, 22 and 25 Januaries (Table 1), not exactly a quarter each. The difference is small next to what I observed (2, 8, 3 and 11 out of 24).
- **A station move.** The station moved 3.2 miles in December 1960 (Part 3). If the new site is naturally a little wetter or drier, part of the change I found could come from the move rather than from the weather. A careful follow-up would redo the analysis using only Januaries after 1960 as the reference period.
- **Scope.** The conclusion applies to this one station and to 1991–2026. It does not say how Idaho as a whole has changed. It also cannot say whether the change comes from climate trends or from natural swings (El Niño/La Niña from year to year, the Pacific Decadal Oscillation from decade to decade).
- **Large sampling fraction.** I sampled 24 of only 36 recent Januaries. The formulas in class assume a very large population, so my test is, if anything, slightly conservative.
- **Class sizes.** The "Wet" class has a small observed count (3), and a couple of different sampled years could change the result. Drawing a second random sample, or using all 36 recent Januaries, would show how stable the conclusion is.

Appendix A: the sample

Table A1. The 24 Januaries drawn by the Simple Random Sample tool (seed 251), in the order they were drawn.

Draw order	year_id	Year	January total (in)	Class
1	13	2003	6.47	Very wet
2	6	1996	4.75	Very wet
3	31	2021	3.98	Very wet
4	5	1995	3.77	Very wet
5	19	2009	3.82	Very wet
6	21	2011	3.83	Very wet
7	27	2017	2.55	Dry
8	7	1997	4.37	Very wet
9	1	1991	2.10	Dry
10	26	2016	4.52	Very wet
11	4	1994	2.42	Dry
12	18	2008	3.06	Wet
13	8	1998	3.51	Wet
14	23	2013	2.09	Dry
15	36	2026	2.08	Dry
16	29	2019	2.37	Dry
17	33	2023	1.45	Very dry
18	30	2020	4.72	Very wet
19	15	2005	1.80	Very dry
20	20	2010	2.93	Wet
21	2	1992	2.67	Dry
22	16	2006	6.87	Very wet

Draw order	year_id	Year	January total (in)	Class
23	14	2004	3.81	Very wet
24	10	2000	2.21	Dry

Appendix B: the historical Januaries

These are the 94 historical January totals I used to set the class cutoffs in Part 2.

Table B1. January total precipitation (in), Moscow U of I station, 1895–1990 (1927 and 1963 not available).

Year: inches	Year: inches	Year: inches	Year: inches	Year: inches	Year: inches
1895: 2.13	1911: 1.00	1928: 3.20	1944: 0.67	1960: 1.22	1977: 0.77
1896: 1.73	1912: 2.77	1929: 3.63	1945: 2.43	1961: 1.85	1978: 3.04
1897: 1.01	1913: 4.32	1930: 1.35	1946: 3.65	1962: 1.01	1979: 1.08
1898: 2.19	1914: 2.68	1931: 2.57	1947: 2.69	1964: 7.25	1980: 3.65
1899: 4.30	1915: 1.36	1932: 3.48	1948: 3.04	1965: 3.48	1981: 1.84
1900: 2.42	1916: 2.67	1933: 5.13	1949: 1.20	1966: 3.34	1982: 3.76
1901: 2.93	1917: 2.86	1934: 2.96	1950: 4.08	1967: 3.46	1983: 2.81
1902: 2.29	1918: 3.21	1935: 2.72	1951: 2.19	1968: 0.96	1984: 2.17
1903: 3.80	1919: 2.26	1936: 5.12	1952: 1.67	1969: 4.83	1985: 0.45
1904: 1.22	1920: 2.79	1937: 3.60	1953: 7.03	1970: 7.67	1986: 4.41
1905: 0.78	1921: 3.95	1938: 1.61	1954: 4.01	1971: 2.89	1987: 1.80
1906: 1.96	1922: 2.04	1939: 1.40	1955: 1.11	1972: 4.30	1988: 2.18
1907: 5.77	1923: 3.93	1940: 2.19	1956: 3.50	1973: 2.57	1989: 3.35
1908: 1.47	1924: 2.34	1941: 1.79	1957: 1.87	1974: 6.70	1990: 4.51
1909: 4.11	1925: 4.07	1942: 1.19	1958: 2.71	1975: 4.96	
1910: 2.58	1926: 2.04	1943: 2.56	1959: 4.77	1976: 1.87	

AI Use Disclosure

> **WHAT A DISCLOSURE LOOKS LIKE**
>
> The project requires you to disclose any AI use. This section is an **illustration of the format**. It is here so you can see what a complete disclosure contains. Write yours about what *you* actually did.

- **Service provider:** Anthropic (Claude).
- **Prompts I used:**
    1. "What are the conditions for a chi-square goodness-of-fit test, and what do I do if an expected count is below 5?"
    2. "Proofread this paragraph for clarity without changing the statistics: [my Part 6 paragraph]."
- **How I used it.** I used the answer to prompt 1 to check the assumptions list in Part 5(B) against my lecture notes, which say the same thing. I used prompt 2 to fix two awkward sentences in Part 6. I did every calculation myself with the stat-tools site and checked each number by hand in Part 5(C). No AI-generated numbers, figures or data appear in this report.
- **Citation:** Anthropic. (2026). *Claude* [Large language model]. https://claude.ai. Responses to prompts 1 and 2 above, accessed September 2026.